SPM 2441 REGRESSION MODELING

Lecture Notes

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Course Details

Course Purpose

To enable the learner to investigate linear relationships between variables using correlation analysis and regression analysis.

Learning Outcomes

At the end of the course the learner should be able to:

1. apply simple linear regression
2. describe and explain multiple linear regression models
3. describe and explain non linear regression models
4. investigate linear relationships between variables using correlation analysis and regression analysis

Course Description

Simple linear regression; least squares formulation and maximum likelihood estimation, tests of hypotheses on slope and intercept. Multiple linear regressions; model description and assumptions, general linear model, least squares estimators and their properties and hypothesis testing in multiple linear regressions. Non-linear regression (generalized linear models); non-linear least squares their estimators and asymptotic properties. Categorical data modeling, (logistic and probit) properties.

Assessment

The course will be assessed as follows:

- CATs - 20%
- Practicals - 10%
- Exam 70%

Software

> Software for use
>
> In this class the R software will be used all through.

To install the R software follow this link to download R.

You will also need to install RStudio an IDE for R follow this link link to download RStudio

Disclaimer

The assumption is that you are knowledgeable enough in using R software. In this class, no R lectures will be offered.

Requirements on Assignments

1. All assignments should be handed in via the Masomo Portal for the respective unit and group.

2. The assignments are to be produced in RMarkdown/Quarto documents either in PDF or MS-Word format.

3. Late submissions will attract a score of 0.

Simple Linear Regression

Introduction

The collection of statistical tools that are used to model and explore relationships between variables that are related in non deterministic manner is called **regression analysis**.

Basics of Regression

We observe a **response** or **dependent** variable Y.

With each Y we also observe **regressors** or **predictors** $\{X_1, X_2, ..., X_n\}$

> Goal
>
> *The goal is to determine the mathematical relationship between response variables and regressors*

The response or dependent variable *varies* with different values of the regressor or predictor.

The predictor values are fixed: we observe the response for these fixed values.

The main focus is in explaining the response variable in association with one or more predictors.

Different forms of regression

There are different forms of regression that exist.

1. Simple Linear Regression
$$Y = \beta_0 + \beta_1 X + \epsilon$$

2. Multiple Linear Regression
$$Y = \beta_0 + \beta_1 X_1 + ... + \beta_p X_p + \epsilon$$

3. Polynomial Regression
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2^2 + \epsilon$$

Simple Linear Regression

The **simple linear regression** is a model with a single regressor x that has a relationship with a response y that is a straight line.

The simple linear regression model is given by:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

where β_0 is the intercept and β_1 is the slope and are unknown constants and ϵ is a random error component.

The errors:

- Assumed to have mean zero and unknown variance σ^2
- The errors are uncorrelated meaning that the value of one error does not depend on the value of the other error.

The response Y is a random variable with the mean of the distribution given as:

$$E(Y|X) = \beta_0 + \beta_1 X$$

and the variance given as

$$Var(Y|X) = Var(\beta_0 + \beta_1 X + \epsilon) = \sigma^2$$

The parameters β_0 and β_1 are usually called the **regression coefficients**.

The slope β_1 is the change in the mean distribution of Y produced by a unit change in X

The intercept β_0 is the mean distribution of the response Y when $X = 0$

Least Squares Estimation of the Parameters

Suppose that we have n pairs of data say $(y_1, x_1), (y_2, x_2), ..., (y_n, x_n)$

Estimation of β_0 and β_1

The **method of least squares** is used to estimate β_0 and β_1.

We may write the regression model as:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, i = 1, 2, ..., n$$

We minimize the squared errors given by:

$$\sum_{i=1}^{n} \epsilon_i^2 = S(\beta_0, \beta_1) = \sum_{i=1}^{n} (Y_i - \beta_0 - \beta_1 X_i)^2$$

We differentiate the function partially wrt to β_0 and β_1 and equate to zero and we have

$$\frac{\partial S}{\partial \beta_0} = -2\sum_{i=1}^{n}(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0$$

$$\frac{\partial S}{\partial \beta_1} = -2\sum_{i=1}^{n}(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)X_i = 0$$

Simplifying these two equations we have:

$$n\hat{\beta}_0 = \hat{\beta}_1\sum_{i=1}^{n}X_i = \sum_{i=1}^{n}Y_i$$

$$\hat{\beta}_0\sum_{i=1}^{n}X_i + \hat{\beta}_1\sum_{i=1}^{n}X_i^2 = \sum_{i=1}^{n}Y_i X_i$$

The above two equations are called **least-squares normal equations** and the solutions to the equations are given as:

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\bar{X} \tag{1}$$

and

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}Y_i X_i - n\bar{x}\bar{y}}{\sum_{i=1}^{n}X_i^2 - n\bar{x}^2} \tag{2}$$

$\hat{\beta}_0$ and $\hat{\beta}_1$ in Equations 1 and 2 are the **least-squares estimators** of the intercept and slope.

The fitted simple linear regression model is then

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X \tag{3}$$

The difference between the observed value Y_i and the corresponding fitted value $\hat{Y}_i$ is the **residual** defined as

$$e_i = Y_i - \hat{Y}_i, i = 1, 2, ..., n \tag{4}$$

Example 1 Consider the data below and fit a simple linear regression model

Table 1: Example 1 data

Y_i	2158.70	1678.15	2316	2061.30	2207.50	1708.30	1784.70	1784.70	2575.00	2357.90
X_i	15.50	23.75	8.00	17.00	5.50	19.00	24.00	2.50	7.50	11.00

Solution

To estimate the model calculate

$$\sum_{i=1}^{n} Y_i X_i = 264028.9, \sum_{i=1}^{n} X_i^2 = 2308.062, \bar{X} = 13.375, \bar{Y} = 2063.225$$

Making the appropriate substitution we have

$$\hat{\beta}_1 = \frac{264028.9 - 10 \times 13.375 \times 2063.225}{2308.062 - 10 \times 13.375^2} = -22.97469$$

$$\hat{\beta}_0 = 2063.225 - (-22.97469 * 13.375) = 2370.51$$

Thus the fitted line is given as:

$$\hat{Y} = 2370.51 - 22.975X \tag{5}$$

One can obtain the fitted values and the residuals of the model using previously defined equations.

Example using R We can fit the model using R and obtain the same results we got by hand.

We start by reading in the data as follows:

```r
# reading in the data
Y <- c(2158.70, 1678.15,2316,2061.30,2207.50,1708.30, 1784.70,1784.70,2575,2357.90)

X <- c(15.50,23.75,8.00,17,5.5,19,24,2.5,7.5,11)
```

We fit the simple linear model and obtain the fitted model results in form of a table

```r
# reading in the data
Y <- c(2158.70, 1678.15,2316,2061.30,2207.50,1708.30, 1784.70,1784.70,2575,2357.90)

X <- c(15.50,23.75,8.00,17,5.5,19,24,2.5,7.5,11)

# fitting the model

model <- lm(Y~X)
knitr::kable(broom::tidy(model))
```

term	estimate	std.error	statistic	p.value
(Intercept)	2370.51090	182.29049	13.004030	0.0000012
X	-22.97465	11.99887	-1.914735	0.0918583

Properties of Least Squares Estimators

We can re-write the estimate of $\widehat{\beta}_1$ as

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i (x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}$$

We note that $\widehat{\beta}_0$ and β_1 are a linear combination of the observations of y_i

Fo instance

$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \sum_{i=1}^{n} c_i y_i$$

where $c_i = \frac{x_i - \bar{x}}{S_{xx}}$

Unbiasedness of the estimators

The least squares estimators $\widehat{\beta}_0$ and β_1 are **unbiased estimators** of the model parameters β_0 and β_1. To show this for $\widehat{\beta}_1$ consider

$$\begin{aligned} E(\widehat{\beta}_1) &= E\Big(\sum_{i=1}^{n} c_i y_i = \sum_{i=1}^{n} c_i E(Y_i)\Big) \\ &= \sum_{i=1}^{n} c_i (\beta_0 + \beta_1 x_i) = \beta_0 \sum_{i=1}^{n} c_i + \beta_1 \sum_{i=1}^{n} c_i x_i \end{aligned}$$

The terms $\sum_{i=1}^{n} c_i = 0$ and $\sum_{i=1}^{n} c_i x_i = 1$

Thus $E[\widehat{\beta}_1] = \beta_1$

Similarly we may show that $\widehat{\beta}_0$ is unbiased estimator of β_0

Consider

$$\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x}$$

$$\begin{aligned} E[\widehat{\beta}_0] &= E[\bar{y} - \widehat{\beta}_1 \bar{x}] \\ &= \bar{y} - \beta_1 \bar{x} \\ &= \beta_0 + \beta_1 \bar{x} - \beta_1 \bar{x} \\ &= \beta_0 \end{aligned}$$